

Quinolines interfere with heme-mediated activation of artemisinin

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This study is **important** as it demonstrates that 4-aminoquinoline antimalarials antagonize artemisinin activity under physiologically relevant conditions. Using isogenic parasite lines and a chemical probe, the authors provide mechanistic insight and **compelling** evidence implicating PfCRT in this antagonism. However, some weaknesses have been identified that limit full interpretation of the findings, which are based solely on in vitro assays, though the results have implications that will be of importance in optimizing future antimalarial combination strategies.

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Abstract

Artemisinin-based combination therapies (ACTs) remain the mainstay of treatment for *Plasmodium falciparum* malaria, despite reports of ACT treatment failure. ACTs consist of an artemisinin and a longer-lived partner drug, which is often a quinoline. Given that heme is central to the mechanism of action of artemisinins and some quinolines, we hypothesized that these antimalarials would exhibit strong drug-drug interactions. Previous studies using standard 48 h or 72 h assays identified additive to mildly antagonistic interactions between artemisinins and quinolines. Here, we sought to re-evaluate these interactions using a pulsing assay that better mimics the short in vivo half-life of artemisinins. We found that chloroquine (CQ), piperaquine (PPQ), and amodiaquine substantially antagonize dihydroartemisinin (DHA), the active metabolite of artemisinins. CQ-DHA antagonism was notably exacerbated in CQ-resistant parasites, resulting in a superantagonistic phenotype in isobolograms. Further, we found that CQ co-treatment conferred artemisinin resistance to Kelch 13 wild type parasites in the ring stage survival assay. Using a small molecule probe to measure chemically reactive heme in live parasites, we determined that quinolines block artemisinin activation by rendering cytosolic heme inert. Finally, we probed beyond traditional ACTs, evaluating interactions of the proposed triple ACT, DHA-PPQ-Mefloquine, as well as OZ439-quinoline combinations, which were all found to be antagonistic. Collectively, these data raise concerns for the clinical use of peroxide-quinoline combination therapies.

Introduction

Malaria was responsible for an estimated 263 million cases and 597,000 deaths in 2023, the majority of which were caused by *Plasmodium falciparum* ¹. The World Health Organization recommends artemisinin-based combination therapies (ACTs) for the treatment of *P. falciparum* malaria. ACTs are composed of a short-lived artemisinin derivative paired with a longer-lived partner drug. Depending on geographical resistance profiles, one of six combinations is recommended: Dihydroartemisinin (DHA) + Piperaquine (PPQ), Artesunate (AS) + Mefloquine (MFQ), AS + Amodiaquine (ADQ), AS + Pyronaridine (PYR), Artemether (AM) + Lumefantrine (LF), or AS + Sulfadoxine-pyrimethamine (SP). Three of these six partner drugs, PPQ, MFQ, and ADQ, possess quinoline groups. Importantly, heme is central to the mechanism of action of both artemisinins and quinolines ^{2,3}.

Asexual blood stage parasites are adept at thriving in an incredibly heme rich environment. Parasites import up to 80% of host cell hemoglobin ⁴, which is transported to a specialized lysosome-like organelle, termed the digestive vacuole. Here, hemoglobin is proteolytically processed into globin peptides and redox active heme (Fe^{2+} -ferroprotoporphyrin IX), which is oxidized to hemin (Fe^{3+} -ferroprotoporphyrin IX) ⁵. The majority of hemin is retained within the digestive vacuole where it is detoxified via sequestration into inert hemozoin crystals ^{6–8}. Hemin concentrations approach 0.5 M within the digestive vacuole ⁹. Blood stage parasites also maintain a relatively high cytosolic heme concentration of approximately 1.6 μM ¹⁰. While the source of this labile heme pool has not been experimentally determined, it is thought to originate from hemoglobin digestion.

The heme-rich environment of asexual blood stages has long been exploited for antimalarial intervention. The 4-aminoquinolines chloroquine (CQ) and PPQ are thought to act primarily in the parasite digestive vacuole, where they inhibit hemozoin formation by binding heme/hemin and the growing face of hemozoin ^{11–16}. It has been proposed that disruption of heme homeostasis and/or accumulation of toxic heme-quinoline complexes may contribute to parasite killing ^{17–19}. Resistance to 4-aminoquinolines is mediated by a collection of mutations in the chloroquine resistance transporter (PfCRT), which is located on the parasite digestive vacuole membrane ²⁰. Mutations in PfCRT are thought to mediate resistance by enabling drug efflux out of the digestive vacuole ^{2,21–24}.

Heme is also central to the mechanism of action of artemisinins. Artemisinins contain a peroxide bridge that must be cleaved by heme for antimalarial activity ^{25–29}. Following activation, artemisinins non-specifically alkylate adjacent molecules, causing widespread cellular damage to the parasite ^{30–32}. Clinically, artemisinin resistance is primarily associated with mutations in Kelch 13 (K13) ^{33–35}, which are thought to reduce parasite hemoglobin uptake and digestion ^{36–38}. Consequently, less heme is available to activate artemisinins.

ACTs remain the mainstay treatment for *falciparum* malaria. Widespread resistance to artemisinins in Southeast Asia and recent emergence of artemisinin resistance in Africa have posed great concern ^{1,34,39–42}. Reduced artemisinin efficacy means that the partner drugs are responsible for clearing a larger parasite biomass, increasing the likelihood for resistance to emerge to partner drugs. Indeed, high treatment failure rates have been reported for AS-MFQ, DHA-PPQ, and AS-ADQ in Southeast Asia ^{43–51}.

Given that heme is central to the mechanism of action of both quinolines and artemisinins, we hypothesized that these antimalarials would exhibit strong drug-drug interactions. Previously, several groups have assessed quinoline-artemisinin interactions using standard 48 or 72 h assays and found that combinations were additive in CQ-sensitive parasites and additive to slightly

antagonistic in CQ-resistant parasites ^{52–55}. However, standard 48 h or 72 h assays in which parasites are exposed to compounds for the entire assay duration do not well represent clinical drug exposure conditions and can mask not only resistance phenotypes, but also drug-drug interaction phenotypes ^{56–59}. We sought to re-evaluate quinoline-artemisinin interactions in multidrug-sensitive, CQ-resistant, artemisinin-resistant, and PPQ-resistant parasites using pulsing assays that better represent artemisinin kinetics. Our work identifies that CQ, PPQ, and ADQ antagonize dihydroartemisinin (DHA), the active metabolite of clinically used artemisinins, by rendering heme chemically inert and preventing DHA activation. In addition, we explore drug-drug interactions beyond traditional ACTs, investigating interactions of putative triple ACTs as well as ozonide-quinoline combinations that failed in clinical trials. Overall, the data presented here highlight the importance of detailed investigation of drug-drug interactions, particularly in drug-resistant parasites.

Results

CQ and DHA are superantagonistic in CQ-resistant parasites

We began by evaluating CQ-DHA interactions in 3D7, a CQ-sensitive parasite, and Dd2, a CQ-resistant parasite (**Table 1**), using trophozoite-stage isobologram assays. Synchronized trophozoites were treated with fixed ratios of CQ and DHA for 4 h, drugs were washed off, and then parasitemia was assessed 72 h later. Given the short half-life of artemisinins ⁵⁷, we believe that this pulsing format allows us to better assess the effect of ACT partner drugs on DHA activity compared to standard 48 or 72 h assays. Fractional IC₅₀ (FIC₅₀) values were calculated from fixed ratios and then plotted on an XY-graph to generate isobolograms. Isobolograms can be interpreted based on isobole shape and mean Σ FIC₅₀ values as follows: points along the dotted line with a mean Σ FIC₅₀ value close to 1 indicate additivity, points above the dotted line in a concave curve with a mean Σ FIC₅₀ value >1.25 indicate antagonism, and points below the dotted line in a convex curve with a mean Σ FIC₅₀ value <0.75 indicate synergy.

We observed strong antagonism for CQ-DHA in 3D7 parasites (**Figure 1a**), with a mean Σ FIC₅₀ value of 1.68. Antagonism was notably exacerbated in Dd2 parasites (**Figure 1b**), which had a mean Σ FIC₅₀ value of 2.44. The magnitude of this value far surpasses conventional antagonism, and we termed this phenomenon “superantagonism”. Perhaps even more unexpected was the atypical shape of this isobole (**Figure 1b**), where points extended upward and peak Σ FIC₅₀ values exceeded 4. In contrast, peak Σ FIC₅₀ values for 3D7 reached only 2. The traditional isobole shape for 3D7 indicates a reciprocal relationship between CQ and DHA, whereas the atypical isobole shape of Dd2 suggests a one-sided relationship, whereby CQ is blocking DHA activity.

3D7 and Dd2 parasites differ genetically at a number of drug resistance markers, including PfCRT which is responsible for mediating CQ resistance. To determine if PfCRT genotype was responsible for differences in CQ-DHA interactions between these parasite lines, we evaluated interactions in two Dd2 parasites that are isogenic except at PfCRT ^{60,61}, Dd2 PfCRT^{3D7}, which is edited to have the 3D7 PfCRT genotype, and Dd2 PfCRT^{Dd2}, which maintains the Dd2 PfCRT genotype and was used as a cloning control (**Table 1**). Notably, superantagonism was lost in Dd2 PfCRT^{3D7} parasites, which phenocopied 3D7 parasites, and had a mean Σ FIC₅₀ value of 1.85 and a conventional isobole shape (**Figure 1c**). In contrast, Dd2 PfCRT^{Dd2} parasites displayed one-sided superantagonism with a mean Σ FIC₅₀ value of 2.73 and peak Σ FIC₅₀ values > 4.5 (**Figure 1d**). These data suggests that PfCRT genotype is the main genetic determinant that dictates CQ-DHA interactions.

Both 3D7 and Dd2 parasites are artemisinin sensitive, harboring a wildtype K13. To determine if resistance to artemisinins alters CQ-DHA interactions, we next evaluated CQ-DHA combinations in Dd2 parasites that are edited to have a K13^{R539T} mutation ³⁵ (**Table 1**). Dd2 K13^{R539T}

Table 1.

Parasite strains used in this study.

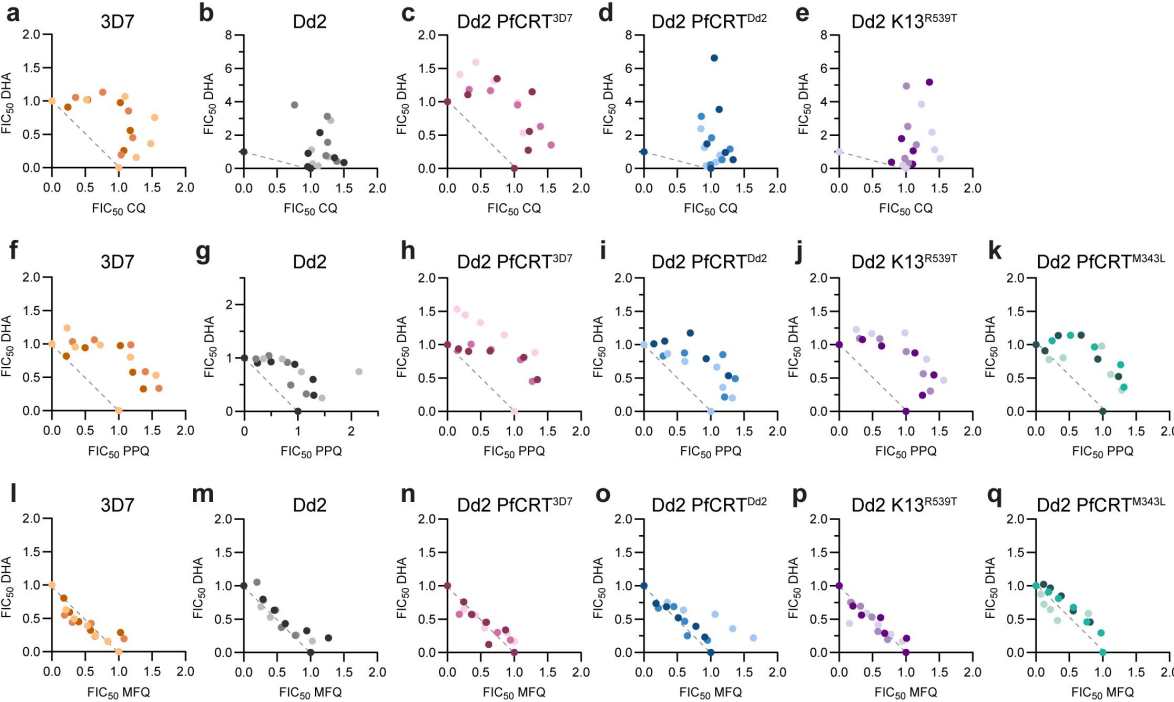
Strains are annotated with their origin (isolate or edited), chloroquine resistance transporter (PfCRT) genotype, Kelch 13 (K13) genotype, and whether they are resistant to CQ, PPQ, or DHA. Wild-type (WT) K13 corresponds to the 3D7 genotype.

Parasite Name	Origin	PfCRT Genotype										K13 Genotype	Resistance
		74	75	76	97	220	271	326	343	356	371		
3D7	Africa (1981)	M	N	K	H	A	Q	N	M	I	R	WT	-
Dd2	Indochina (1980)	I	E	T	H	S	E	S	M	T	I	WT	CQ
Dd2 PfCRT ^{Dd2}	Edited (introns removed)	I	E	T	H	S	E	S	M	T	I	WT	CQ
Dd2 PfCRT ^{3D7}	Edited	M	N	K	H	A	Q	N	M	I	R	WT	-
Dd2 K13 ^{R539T}	Edited	I	E	T	H	S	E	S	M	T	I	R539T	CQ, DHA
Dd2 PfCRT ^{M343L}	Edited	I	E	T	H	S	E	S	L	T	I	WT	CQ, PPQ
MRA-1284 (IPC_6261)	Cambodia (2012)	I	E	T	Y	S	E	S	M	T	I	C580Y	DHA, PPQ

Figure 1.

CQ and DHA are superantagonistic in CQ-resistant parasites.

Shown are trophozoite-stage isobolograms for (a-e) CQ-DHA, (f-k) PPQ-DHA, and (l-q) MFQ-DHA. The dotted line on each graph represents perfect additivity. 3D7, Dd2, Dd2 PfCRT^{3D7}, Dd2 PfCRT^{Dd2}, Dd2 K13^{R539T}, and Dd2 PfCRT^{M343L} parasites are indicated in orange, grey, pink, blue, purple, and teal, respectively. Data from three independent replicates are indicated by different shading. Figure supplement 1. CQ, DHA, sity of different parasites.



parasites, which also harbor a Dd2 PfCRT genotype, displayed one-sided superantagonistic isobole that was reminiscent of both Dd2 and Dd2 PfCRT^{Dd2} parasites (**Figure 1e**). This suggests that, unlike PfCRT genotype, K13 genotype appeared to have no impact on drug-drug interactions at the trophozoite stage.

PPQ and DHA are antagonistic

Next, we sought to evaluate drug-drug interactions of DHA and the ACT partner drugs PPQ and MFQ. PPQ is believed to have a similar mechanism of action as CQ and both antimalarials are 4-aminoquinolines. Accordingly, we hypothesized that PPQ and DHA would also be antagonistic. In 3D7, Dd2, Dd2 PfCRT^{3D7}, Dd2 PfCRT^{Dd2}, and Dd2 K13^{R539T} trophozoites, we observed conventional antagonism for PPQ-DHA (**Figure 1f-j**) with mean Σ FIC50 values ranging from 1.55 to 1.71. To determine if, similar to CQ, PPQ-DHA interactions would be exacerbated in PPQ-resistant parasites, we also evaluated Dd2 PfCRT^{M343L}, which harbors the Dd2 PfCRT plus an additional mutation at M343L⁶¹ (**Table 1**). The M343L mutation had no impact on PPQ-DHA interactions and Dd2 PfCRT^{M343L} parasites displayed conventional antagonism with a mean Σ FIC50 value of 1.58 (**Figure 1k**). Note that while the PPQ IC50 of Dd2 PfCRT^{M343L} is 2.5-fold higher than Dd2 (65 nM versus 25 nM; Figure 1-figure supplement 1), this shift is substantially smaller than the >1000-fold CQ IC50 shift that we and others⁵⁹ have observed in CQ-sensitive versus CQ-resistant parasites using the pulsing assay format (Figure 1-figure supplement 1).

Next, we assessed interactions between MFQ and DHA (**Figure 1l-q**). Though MFQ contains a quinoline group, it is chemically classified as an aryl amino alcohol and is believed to have a heme-independent mechanism of action⁶². Consistent with this, we observed additivity for MFQ-DHA in all six parasites tested with Σ FIC50 values ranging from 0.94 to 1.20. Overall, these data indicate that CQ and PPQ are antagonistic with DHA, while MFQ is additive.

CQ combination treatment confers artemisinin resistance

to K13^{WT} parasites in ring-stage survival assays

Given our observation that CQ and PPQ antagonize DHA in trophozoite stages, we next sought to determine if these quinolines would promote survival in early ring stages where artemisinin resistance is classically observed. We performed DHA dose-response assays on early ring stage parasites to determine both ring-stage survival assay (RSA) values and IC50 values (**Figure 2a** and Figure 2-figure supplement 1). The RSA is used to delineate artemisinin-sensitive versus resistant parasites, where survival > 1% corresponds to resistance and < 1% indicates sensitivity⁵⁷. Though DHA IC50 values cannot reliably differentiate artemisinin sensitive vs resistant parasites⁵⁷, they can provide important insight into different factors that contribute to artemisinin susceptibility⁶³.

In the absence of quinolines, the artemisinin sensitive K13^{WT} parasites, Dd2 PfCRT^{3D7} and Dd2, had RSA survival values $\leq$ 1%, while Dd2 K13^{R539T} had an RSA survival of 12.63% (**Figure 2b-d top panel**). Co-treatment with 10 μ M CQ, a pharmacologically relevant concentration^{64,65}, significantly increased RSA survival, conferring artemisinin resistance to both Dd2 PfCRT^{3D7} and Dd2 parasites with mean RSA values of 2.66% and 4.49%, respectively (**Figure 2b and c top panel**). RSA survival values were also increased in Dd2 K13^{R539T} parasites to 18.39%, although this shift was not statistically significant (**Figure 2d top panel**). When assessing DHA sensitivity by ring-stage IC50 values, CQ co-treatment also had a substantial rescuing effect in both Dd2 and Dd2 K13^{R539T} parasites, resulting in a 4-fold increase in DHA IC50 (**Figure 2c and d bottom panel**). Note that while we were unable to determine a DHA IC50 value for CQ-sensitive Dd2 PfCRT^{3D7} parasites in the presence of 10 μ M CQ, the dose-response curve was suggestive of antagonism (Figure 2-figure supplement 1a).

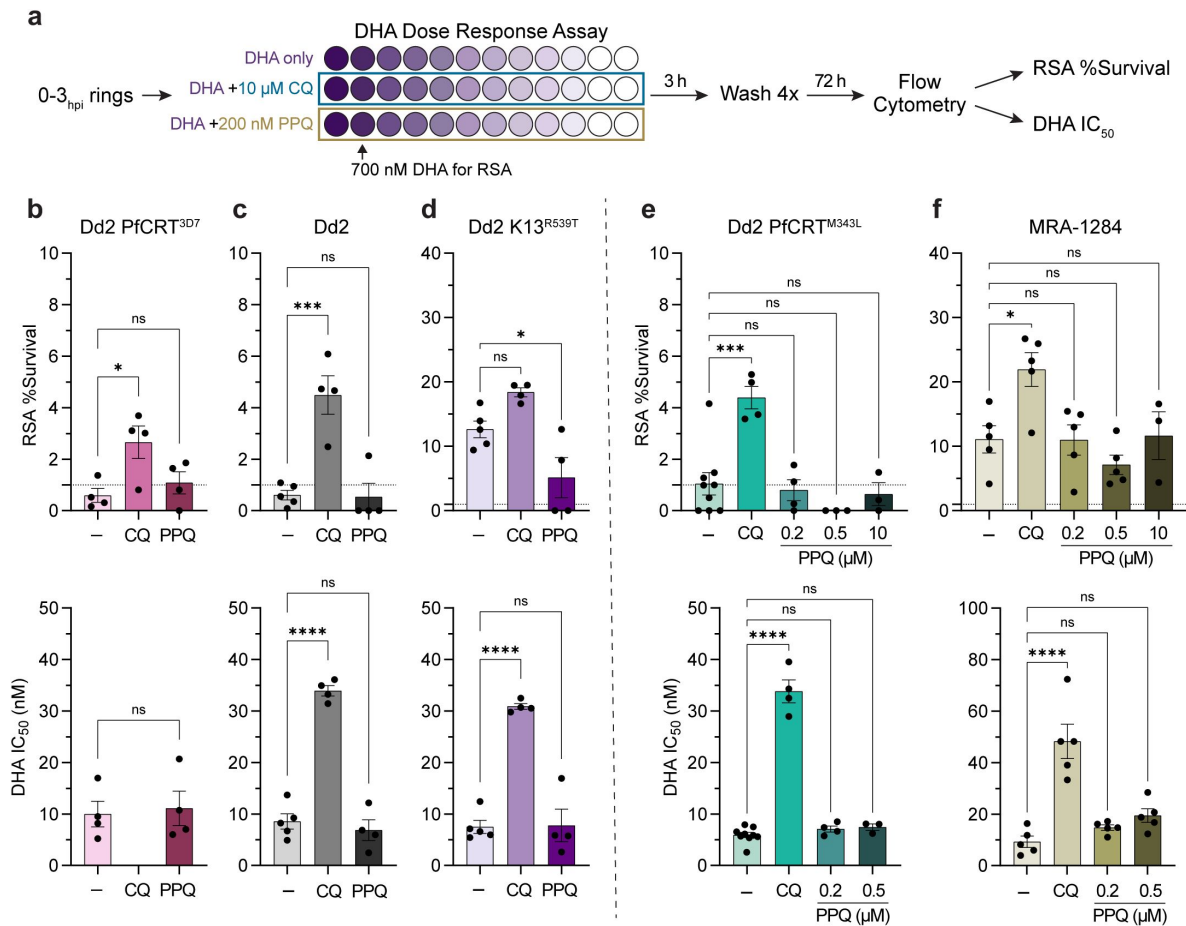


Figure 2.

CQ co-treatment confers artemisinin resistance in K13^{WT} parasites.

(a) DHA dose response assays were performed on 0-3 hours post invasion (hpi) ring stage parasites. Two-fold serial dilutions of DHA were prepared in 96-well plates (DHA concentration represented by different shades of purple) starting at 1.4 μM so that both IC₅₀ values and RSA survival values at 700 nM DHA could be determined. To evaluate quinoline-DHA interactions in early ring stages, DHA titrations were prepared with fixed concentrations of 10 μM CQ or 200 nM PPQ. (b-d) Mean RSA survival values ± SEM (**top panel**) and mean DHA IC₅₀ values ± SEM (**bottom panel**) are shown for (b) Dd2 PfCRT^{3D7}, (c) Dd2, and (d) Dd2 K13^{R539T} parasites. For the PPQ-resistant parasites (e) Dd2 PfCRT^{M343L} and (f) MRA-1284, additional fixed PPQ concentrations of 500 nM and 10 μM were also tested. For **b bottom panel**, a student's t-test was used to assess statistical differences between DHA alone versus DHA + PPQ. For all other graphs, statistical differences between DHA alone (-) and DHA + quinolines was assessed using a one-way ANOVA with a Dunnett's test for multiple comparisons. *****p* < 0.0001; ****p* < 0.001; **p* < 0.05; ns = not significant. Figure supplement 1. Early ring stage dose-response curves.

To determine if PPQ would have a similar rescuing effect, we performed these same assays in the presence of 200 nM PPQ. This concentration was chosen because it is sublethal for early rings in this short 3 h pulse, pharmacologically relevant ^{66,67}, and mimics the PPQ survival assay (PSA) that is used to delineate PPQ resistance ⁵⁶. Co-treatment with 200 nM PPQ did not have a rescuing effect for Dd2 PfCRT^{3D7}, Dd2, or Dd2 K13^{R539T} parasites and resulted in mean RSA and IC50 values that were either similar to or lower than the DHA-only control (**Figure 2b-d**). Notably, co-treatment with 200 nM PPQ did not shift the DHA dose response curves of these parasites (Figure 2-figure supplement 1a-c), which is consistent with a conventional antagonistic relationship.

We were curious if higher concentrations of PPQ might have a rescuing effect, particularly for PPQ-resistant parasites that are capable of surviving a higher PPQ dose. To this end, we assayed two PPQ-resistant parasites: Dd2 PfCRT^{M343L} and MRA-1284 (IPC_6261), which is an artemisinin-resistant and PPQ-resistant clinical isolate that was collected in 2012 from Cambodia (**Table 1**). Dd2 PfCRT^{M343L} and MRA-1284 have PPQ survival values (PSA) of 10% ⁶¹ and 40% ⁶⁸, respectively. While co-treatment with 10 μ M CQ had a rescuing effect for Dd2 PfCRT^{M343L} as assessed by RSA and IC50 values, neither 200 nM PPQ, 500 nM PPQ, or 10 μ M PPQ co-treatment rescued parasite survival (**Figure 2e** and Figure 2-figure supplement 1d). Similarly, 10 μ M CQ co-treatment decreased MRA-1284 sensitivity to DHA as assessed by RSA and early ring IC50 values, while co-treatment with 200 nM PPQ or 500 nM PPQ had no effect on DHA sensitivity (**Figure 2f**). Interestingly, while co-treatment with 10 μ M PPQ did not alter RSA survival of MRA-1284, it did shift the dose-response curve to the left (Figure 2-figure supplement 1e). However, since 10 μ M PPQ inhibited parasite growth approximately 60% on its own, we were unable to determine a IC50 value.

CQ and PPQ antagonize DHA activation by rendering heme inert

In total, we profiled three different quinoline-DHA interactions, which are summarized in **Figure 3a**. In solution, quinolines and heme form heme-quinoline complexes, wherein heme is present as a μ -oxo dimer ^{69–71}. Though the exact stoichiometry of this interaction for each quinoline remains debated ^{69–73}, formation of such a complex could render quinoline-bound heme chemically inert (i.e. unable to activate artemisinins).

To investigate this, we utilized a recently developed small molecule probe, H-FluNox, that is highly sensitive and specific for heme over hemin, heme-bound proteins, Fe²⁺, and other salts and metal ions ⁷⁴. H-FluNox is basally non-fluorescent, but upon cleavage of a N-O bond by the Fe²⁺ of heme, H-FluNox fluoresces ⁷⁴ (**Figure 3b top panel**). Importantly, this reaction is similar to artemisinin activation, which requires cleavage of artemisinin's peroxide bond by the Fe²⁺ of heme ²⁹ (**Figure 3b bottom panel**). To determine if quinoline-heme complexes would react with H-FluNox, we began by assessing H-FluNox fluorescence in solution with 10 μ M CQ, which is sub-lethal for CQ-resistant parasites and pharmacologically relevant ^{64,65}. Relative to the untreated control, 10 μ M CQ inhibited fluorescence approximately 50% in the presence of 250 nM heme (**Figure 3c**). This suggests that CQ-heme complexes have diminished capacity to activate artemisinins. Next, we evaluated PPQ and MFQ at two different concentrations: 150 nM, which reflects a 5x trophozoite-stage IC50 concentration, and 10 μ M to directly compare these quinolines with CQ. At 150 nM, both PPQ and MFQ inhibited fluorescence, albeit only 10% (**Figure 3c**). At 10 μ M, MFQ inhibited fluorescence to a similar extent as CQ, while 10 μ M PPQ only inhibited fluorescence 25% (**Figure 3c**). Note that 10 μ M concentrations exceed maximum plasma concentrations of PPQ and MFQ, which have been reported to reach approximately 0.5-1 μ M and 5 μ M, respectively ^{66,67,75}.

Next, we assessed fluorescence in live trophozoite stage parasites using the cell permeable analog of H-FluNox, Ac-H-FluNox ⁷⁴. In untreated Dd2 parasites, fluorescence was predominantly observed in the parasite cytoplasm (**Figure 3d**), and little signal was detected in red blood cells or the parasite digestive vacuole. This suggests that a significant portion of DHA could be activated

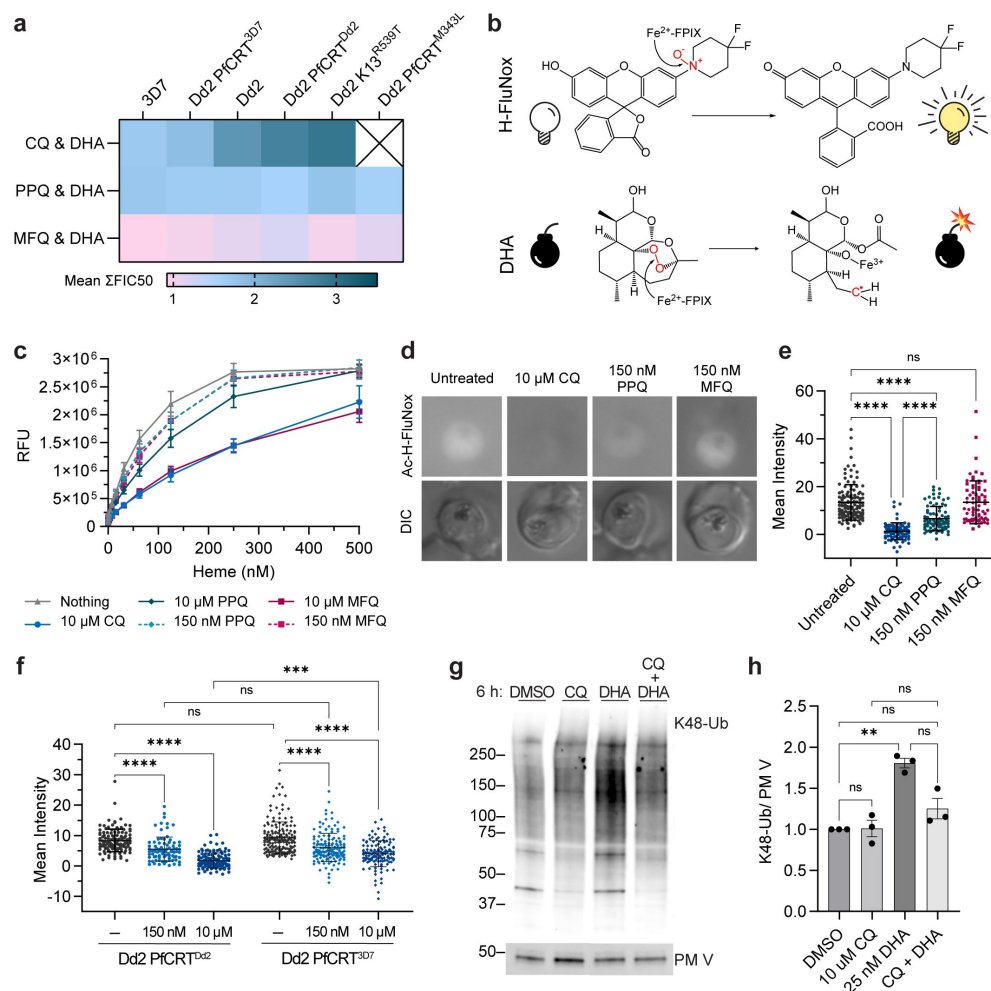


Figure 3.

CQ and PPQ antagonize DHA activation by “inactivating” heme.

(a) Shown are mean Σ FIC50 values calculated from the trophozoite stage isobolograms in [Figure 1](#). Values close to 1 (pink) indicate additivity. Values ranging from 1.25 to 2 (light blue) indicate classical antagonism. Values > 2.25 (dark turquoise) indicate superantagonism. **(b top panel)** Heme (Fe^{2+} -FPIX) cleaves the N-O bond of H-FluNox, mediating probe fluorescence. **(b bottom panel)** Heme cleaves the peroxide bridge of DHA, which generates a carbon centered radical that can alkylate proximal biomolecules. **(c)** H-FluNox was incubated with increasing concentrations of heme in the presence or absence of 10 μM CQ, 150 nM PPQ, 10 μM PPQ, 150 nM MFQ, or 10 μM MFQ. Shown are mean relative fluorescence units (RFU) $\pm$ SD from at least three independent replicates performed in technical triplicate. **(d and e)** Dd2 trophozoites were treated with 10 μM CQ, 150 nM PPQ, 150 nM MFQ, or mock treated for 5.5 h. **(f)** Dd2 PfCRT^{Dd2} and Dd2 PfCRT^{3D7} trophozoites were mock treated or treated with 150 nM CQ or 10 μM CQ for 5.5 h. **(d-f)** Parasites were then incubated with 10 μM Ac-H-FluNox (the cell permeable analog of H-FluNox) to visualize and quantify “active” heme. Shown are **(d)** representative images and **(e and f)** mean fluorescence intensity $\pm$ SD of at least 70 parasites from at least three independent drug treatments. Statistical significance was assessed using a one-way ANOVA with a Turkey’s test for multiple comparisons. **(g and h)** Dd2 trophozoites were treated with 10 μM CQ, 25 nM DHA, 10 μM CQ + 25 nM DHA for 6 h. Parasite lysates were subjected to western blot and probed with anti-K48-linked ubiquitin (K48-Ub) antibodies or anti-plasmeepsin V (PMV) antibodies. Shown is a **(g)** representative blot and **(h)** quantification of relative K48-Ub intensity $\pm$ SEM from three independent replicates. Statistical significance between mock treated and antimalarial-treated parasites was determined using a one-way ANOVA with a Turkey’s test for multiple comparisons. **** $p < 0.0001$; *** $p < 0.001$; ** $p < 0.01$; ns = not significant. Figure supplement 1. Effect of pH on H-FluNox and Ac-H-FluNox activity. Figure supplement 2. Equipolar comparison of CQ, PPQ, and MFQ on Ac-H-FluNox fluorescence. Figure supplement 3. Representative images of Dd2 PfCRT^{Dd2} and Dd2 PfCRT^{3D7} Ac-H-FluNox fluorescence. Figure supplement 4. Additional K48-ubiquitin western blots. Table supplement 1. Mean parasite intensity for Ac-H-FluNox experiments. Source data 1. Uncropped western blots used for ImageJ quantification.

in the parasite cytoplasm. Lack of signal in the digestive vacuole could be due to issues with permeability, the specificity of Ac-H-FluNox for heme over hemin (the predominant heme species in the digestive vacuole), or sensitivity of Ac-H-FluNox to a low pH environment. In solution, H-FluNox fluorescence was decreased approximately 60% at pH 5.4 (Figure 3-figure supplement 1a), which is the estimated pH of the digestive vacuole ⁷⁶. However, alkalization of the digestive vacuole with ammonium chloride ⁷⁷ did not increase digestive vacuole fluorescence signal (Figure 3-figure supplement 1b), suggesting that low pH alone cannot explain this lack of labeling.

Importantly, treatment of Dd2 trophozoites with 10 μ M CQ nearly ablated Ac-H-FluNox fluorescent signal (Figure 3d and e), suggesting that CQ-heme complexes are also unable to activate artemisinins in parasites. Treatment with 150 nM PPQ significantly reduced fluorescence, though not to the same extent as 10 μ M CQ (50% inhibition versus > 92% inhibition) (Figure 3d, e, and Figure 3 and 4-table supplement 1). In contrast, parasites treated with 150 nM MFQ had similar fluorescence as untreated control parasites (Figure 3d and e). These assays with H-FluNox and Ac-H-FluNox correlate well with hemozoin inhibition assays in which CQ, PPQ, and MFQ inhibit hemozoin formation in solution ⁷⁰, but only CQ and PPQ inhibit hemozoin formation in parasites ^{73,78–80}. To determine if concentration alone was responsible for the differences that we observed between CQ and the other quinolines, we also evaluated Ac-H-FluNox fluorescence following treatment with 10 μ M PPQ or 10 μ M MFQ. At this super-physiological and super-lethal concentration, an equimolar concentration of CQ still inhibited H-FluNox fluorescence to a greater extent than PPQ or MFQ (Figure 3-figure supplement 2 and Figure 3 and 4-table supplement 1). This suggests that something unique to the chemistry of CQ and/or the biology of the parasite is responsible for the differences that we observed.

PfCRT^{Dd2} is believed to mediate CQ resistance by pumping CQ out of the digestive vacuole ^{23,21–24}, suggesting that cytoplasmic CQ concentrations may differ between PfCRT^{Dd2} and PfCRT^{3D7} parasites treated with the same concentration of CQ. Given the substantial difference that we saw in the shape of CQ-DHA isobolograms between PfCRT^{Dd2} and PfCRT^{3D7} parasites, we were curious if differences in PfCRT genotype would impact Ac-H-FluNox fluorescence. Accordingly, we treated Dd2 PfCRT^{Dd2} and Dd2 PfCRT^{3D7} parasites with 150 nM CQ or 10 μ M CQ and measured chemically “reactive” heme using Ac-H-FluNox. In untreated Dd2 PfCRT^{Dd2} versus Dd2 PfCRT^{3D7} parasites, we observed no difference in Ac-H-FluNox localization or fluorescence intensity (Figure 3f and Figure 3-figure supplement 3). Following treatment with 150 nM CQ, fluorescence was inhibited 25% in Dd2 PfCRT^{3D7} and 35% in Dd2 PfCRT^{Dd2}, although this difference was not statistically significant (Figure 3f and Figure 3 and 4-table supplement 1). Curiously, while 10 μ M CQ treatment reduced mean fluorescence approximately 60% in Dd2 PfCRT^{3D7} parasites, it was not to the extent of Dd2 PfCRT^{Dd2} parasites (80%) (Figure 3f and Figure 3 and 4-table supplement 1). These data suggest that transport of CQ out of the digestive vacuole by PfCRT^{Dd2} may contribute to superantagonism between CQ and DHA in CQ-resistant parasites.

CQ co-treatment rescues DHA-mediated protein damage

Following activation by heme, DHA causes widespread protein damage in the parasite ^{30–32,81}. Accordingly, we would expect that if CQ is antagonizing DHA activation, then CQ co-treatment should rescue DHA-mediated protein damage. To test this hypothesis, we treated Dd2 trophozoites with CQ, DHA, or CQ + DHA and probed for K48-linked ubiquitin, which is a marker of protein damage ⁸¹. Consistent with what has been previously reported ⁸², CQ treatment alone had no effect on K48-linked ubiquitin levels, while treatment with DHA resulted in a 2-fold increase in K48-linked ubiquitin (Figure 3g, h, and Figure 3-figure supplement 4). Importantly, addition of CQ to DHA had a rescuing effect, resulting in levels of K48-linked ubiquitin that were similar to the DMSO-treated control (Figure 3g and h). Collectively, these data suggest that CQ antagonizes DHA by rendering heme chemically inert, reducing DHA activation, and consequently, reducing DHA-mediated protein damage.

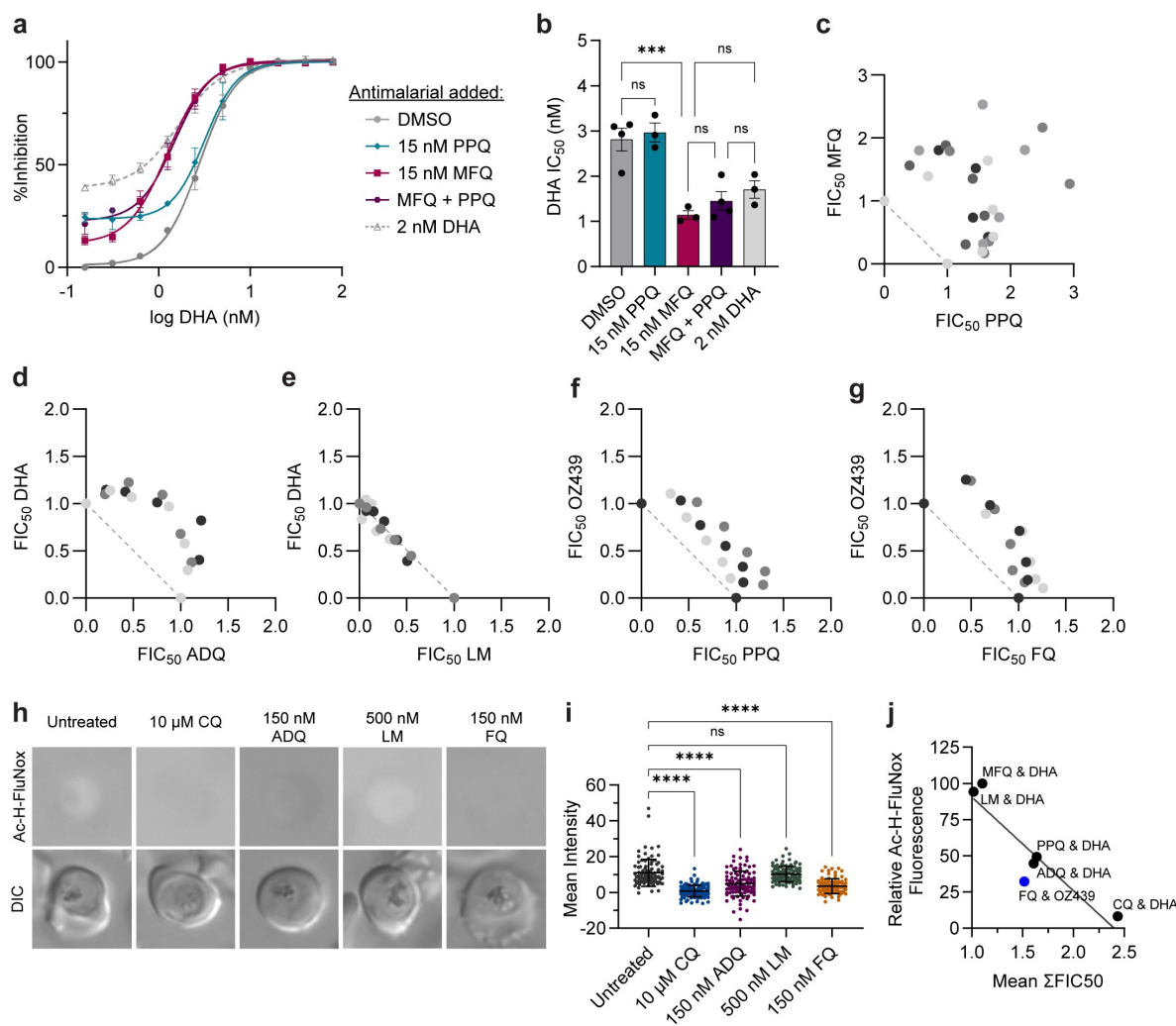


Figure 4.

Quinolines antagonize peroxides.

(a and b) Trophozoite-stage DHA dose-response assays were performed with Dd2 parasites in the presence or absence of 15 nM PPQ, 15 nM MFQ, 15 nM PPQ + 15 nM MFQ, or 2 nM DHA. Shown are (a) dose-response curves with mean % inhibition $\pm$ SEM and (b) mean IC₅₀ values $\pm$ SEM from at least three independent replicates. Statistical significance was determined using a one-way ANOVA with a Turkey's test for multiple comparisons. Trophozoite-stage isobolograms were performed on Dd2 parasites to determine drug-drug interactions of (c) PPQ-MFQ, (d) ADQ-DHA, (e) LM-DHA, (f) PPQ-OZ439, and (g) FQ-OZ439 using the following fixed ratios: 1:0, 4:1, 2:1, 1:1, 1:2, 1:4, 0:1. Shown are fractional IC₅₀ values from at least 3 independent replicates. Independent replicates are indicated in different shading and the diagonal dotted line on each plot indicates perfect additivity. (h and i) Dd2 parasites were mock treated or treated with 10 μM CQ, 150 nM ADQ, 500 nM LM, or 150 nM FQ for 5.5 h and free heme was then labeled in live parasites with Ac-H-FluNox. Shown are (h) representative images and (i) mean fluorescence intensity $\pm$ SD of at least 90 parasites from three independent drug treatments. Statistical significance was assessed using a one-way ANOVA with a Dunnett's test for multiple comparisons. **** $p < 0.0001$; ns = not significant. (j) Mean relative fluorescence of Ac-H-FluNox from drug treatments in Figure 3e and 4i was plotted against Dd2 mean ΣFIC₅₀ values for the indicated combinations. Combinations with DHA are indicated in black. Combinations with OZ439 are indicated in blue. Figure supplement 1. Additional MFQ isobolograms. Figure supplement 2. ADQ, LM, and FQ IC₅₀ values. Table supplement 1. Mean parasite intensity for Ac-H-FluNox experiments.

PPQ and MFQ do not harm treatment efficacy, but strongly antagonize each other

With the looming threat of widespread ACT treatment failure, several ideas have been proposed to delay the spread of ACT resistance by using currently available antimalarials. Triple ACTs (TACTs), which are composed of a traditional ACT plus an additional longer-lived partner drug, have gained the most traction. Early clinical trials indicate that DHA-PPQ-MFQ and AM-LM-ADQ show good safety and efficacy against artemisinin-sensitive and resistant parasites^{47,83–85}. We were curious how the addition of a third antimalarial would influence drug interactions. Accordingly, we performed trophozoite-stage DHA dose response assays on Dd2 parasites in the presence of a fixed, sub-lethal concentration of PPQ, MFQ, or PPQ + MFQ. Addition of 2 nM DHA was also included as a control to represent additivity. As assessed by DHA IC₅₀ values, addition of 15 nM PPQ did not improve parasite killing, while addition of 15 nM MFQ resulted in a 3-fold decrease in IC₅₀ compared to the DMSO control (**Figure 4a and b**). These phenotypes are consistent with antagonistic and additive relationships, respectively. Notably, addition of PPQ + MFQ had no additional benefit compared to addition of MFQ alone (**Figure 4a and b**).

Following the initial 3-day TACT treatment regimen, long-lived partner drugs are responsible for clearing the remaining parasite burden. Accordingly, we sought to assess drug-drug interactions of PPQ-MFQ in the absence of DHA. Previously it was reported that MFQ and PPQ were moderately antagonistic in 3D7 and K1 parasites using a standard 48 h assay, with mean Σ FIC₅₀ values of 1.31 and 1.28, respectively⁵². Using our trophozoite-stage assay, we observed reciprocal superantagonism in both PPQ-sensitive, Dd2 (**Figure 4c**), and PPQ-resistant Dd2 PfCRT^{M343L} (Figure 4-figure supplement 1a), with mean Σ FIC₅₀ values of 2.62 and 2.44, respectively. This phenotype appears to be unique to PPQ, as we observed only strong antagonism for CQ-MFQ with a mean Σ FIC₅₀ of 1.72 (Figure 4-figure supplement 1b). Overall, these data suggest that while adding PPQ and MFQ to DHA does not harm treatment efficacy, PPQ and MFQ may not be a strategic antimalarial pairing for TACTs given their superantagonism with each other.

Quinolines antagonize peroxides

To determine if the drug-drug interactions that we evaluated were specific to a subset of antimalarials or represented a broader phenotype, we next evaluated DHA interactions with the clinically used ACT partner drugs ADQ and LM. Structurally, ADQ is classified as a 4-aminoquinoline like PPQ and CQ, while LM is classified as an aryl amino alcohol like MFQ. However, unlike MFQ, LM lacks a quinoline group. We found that in Dd2 parasites, ADQ and DHA were antagonistic (**Figure 4d**) with a mean Σ FIC₅₀ of 1.61, while LM and DHA were additive (**Figure 4e**) with a mean Σ FIC₅₀ of 1.02.

To determine if 4-aminoquinolines antagonize other peroxide antimalarials, we next sought to assess interactions of two different OZ439-quinoline pairs that were previously evaluated in clinical trials. Similar to artemisinins, the synthetic ozonide, OZ439, requires activation via heme-mediated cleavage of a peroxide bond. In phase II clinical trials, OZ439 showed good efficacy alone⁸⁶, but poor efficacy either in combination with PPQ⁸⁷ or the next generation quinoline, ferroquine (FQ)^{88,89}. To assess drug-drug interactions, we performed trophozoite-stage isobolograms with OZ439-PPQ and OZ439-FQ in Dd2 parasites. Both PPQ-OZ439 (**Figure 4f**) and FQ-OZ439 (**Figure 4g**) were antagonistic with a mean Σ FIC₅₀ values of 1.42 and 1.52, respectively. These data suggest that antagonistic interactions between OZ439 and quinolines may, in part, have contributed to the poor efficacy of these single dose combination therapies.

Although the mechanism of action of ADQ and FQ are less well studied, both antimalarials appear to inhibit hemozoin formation. ADQ was shown to inhibit hemozoin formation in parasites using the heme fractionation assay⁸⁰. While in-parasite assays have not been performed with FQ, in vitro biochemical studies suggest that FQ may inhibit hemozoin formation even better than CQ

⁹⁰ Accordingly, we were curious if ADQ and FQ treatment would “inactivate” heme in parasites. Using the Ac-H-FluNox heme probe, we observed that relative to the untreated control, treatment with 150 nM ADQ or 150 nM FQ (approximate 5x IC₅₀ concentrations; Figure 4-figure supplement 2) reduced Ac-H-FluNox fluorescent signal approximately 50% and 55%, respectively (Figure 4h, i, and Figure 3 and 4-table supplement 1). In contrast, treatment with an equipotent concentration of LM had no significant impact on Ac-H-FluNox signal relative to the untreated control (Figure 4h and i). Among the six partner drugs that were evaluated, we observed a clear inverse correlation between Ac-H-FluNox signal and mean Σ FIC₅₀, with an R² value of 0.84 (Figure 4j). These data strongly suggest that antimalarials that render heme inert also antagonize artemisinin and ozonide antimalarials, presumably by blocking endoperoxide activation.

Discussion

Combination therapies are widely believed to delay the emergence of drug resistance not only for malaria, but also bacterial infections, HIV, fungal infections, and cancer. However, to maximize the benefits of combination therapies, it is essential that drugs are strategically paired. At a minimum, combinations should enhance efficacy to reduce parasite biomass and reduce the potential for drug resistance to evolve. In vitro assessment of drug-drug interactions between two antimalarials with vastly different half-lives is challenging. We chose to use a short treatment pulse to mimic artemisinin exposure and gain an understanding into how quinolines impact artemisinin activity. While drug-drug interactions are undoubtedly much more complex in vivo, we believe this assay format better recapitulates clinical conditions compared to traditional 48 or 72 h assays.

We found that CQ, PPQ, and ADQ are antagonistic with DHA as assessed in early ring and trophozoite stages. Our work suggests that these antimalarials antagonize DHA activation by rendering intra-parasitic heme chemically inert. To date, it remains debated where artemisinins are activated within the parasite. There are several lines of evidences that suggest that artemisinins can be activated in the parasite cytoplasm. First, immunofluorescence studies with an anti-AS antibody⁹¹ and live cell imaging with an artemisinin-based photoaffinity probe⁹² revealed that artemisinins localize throughout the parasite cytoplasm. Second, numerous chemical proteomics studies utilizing artemisinin and ozonide-based click chemistry probes have identified a number of cytoplasmic proteins as targets of artemisinin and ozonides^{30–32,93,94}. Given how quickly activated artemisinins react with adjacent molecules, it is unlikely that activated artemisinins could traverse the digestive vacuole or other organelle membranes before encountering a cytoplasmic target. Finally, immunofluorescence assays with antibodies specific for the alkylation signature of the ozonide OZ277⁹⁵ revealed labeling throughout the parasite cytoplasm. Our data with Ac-H-FluNox also lend credence to this hypothesis. We observed Ac-H-FluNox fluorescence throughout the parasite cytoplasm, indicating that there is a substantial amount of heme in this location that is chemically competent to activate artemisinins. It is possible that technical limitations, such as issues with membrane permeability, limit Ac-H-FluNox fluorescence in the digestive vacuole. Nevertheless, these data collectively indicate that artemisinins can be activated in the parasite cytoplasm.

Previously, it has been reported that CQ treatment increases intracellular “free” heme^{7,10,96}. However, the term “free” heme is somewhat malleable. While the pyridine-heme fractionation assay^{7,80}, electron spectroscopic imaging (ESI) with energy-loss spectroscopy (EELS)⁷ and heme biosensor¹⁰, can be used to detect and quantify heme species, they are unable to differentiate chemically “active” heme from inert heme (i.e. quinoline bound or protein bound). In contrast, H-FluNox is highly specific for chemically “active” heme⁷⁴. One possibility to reconcile our data with previous reports is that CQ treatment increases non-hemozoin/hemoglobin heme, but that this heme is bound to CQ or other biomolecules, rendering it

unable to activate peroxides. Importantly, activation of Ac-H-FluNox is very similar to that of artemisinins, providing a useful proxy to assess not only chemically active heme, but also artemisinin activation in live parasites.

CQ-DHA antagonism was notably exacerbated in CQ-resistant parasites. The concentrations used in isobolograms are chosen based on parasite IC₅₀ values to individual compounds. While all parasites were exposed to similar DHA concentrations, CQ-resistant parasites were exposed to much higher CQ concentrations compared to their CQ-sensitive counterparts. This may explain in part why antagonism was intensified in CQ resistant parasites. However, equimolar treatment with 10 μ M CQ resulted in significantly lower Ac-H-FluNox signal in Dd2 PfCRT^{Dd2} versus Dd2 PfCRT^{3D7} parasites (**Figure 3f**). These data could suggest that while the majority of CQ is sequestered in the digestive vacuole of Dd2 PfCRT^{3D7} parasites, PfCRT^{Dd2} is able to transport more CQ out of the digestive vacuole, thus allowing more CQ to bind cytosolic heme in parasites harboring a PfCRT^{Dd2}. This duality of PfCRT^{Dd2}, simultaneously enhancing parasite survival to CQ while promoting cytoplasmic heme “inactivation”, may explain the unprecedented one-sided superantagonistic isobole that we observed.

Fortunately, we did not observe superantagonism with DHA-PPQ in PPQ-resistant parasites. With a modest 3-fold shift in PPQ IC₅₀ between Dd2 vs Dd2 PfCRT^{M343L} parasites (25 nM vs 65 nM), it is perhaps unsurprising that no significant difference in DHA-PPQ isobolograms was observed between these two parasites. This is in contrast to the >1000-fold CQ IC₅₀ shift that we and others⁵⁹ have observed between CQ-sensitive and CQ-resistant parasites using pulsing assays. Our studies with Ac-H-FluNox indicate that 150 nM PPQ is sufficient to render approximately 50% of available heme inert (**Figure 3e**). While these data suggest that PPQ blocks some DHA activation, it is likely that 150 nM PPQ is not sufficient to block all DHA activation. Since even PPQ-resistant parasites are at least partially susceptible to mid-nanomolar PPQ concentrations, the protective effect that PPQ has in neutralizing DHA is partially negated by PPQ-mediated killing. In contrast, CQ-resistant parasites are relatively unaffected by pulses with low micromolar CQ concentrations. Accordingly, CQ concentrations that are high enough to nearly block all DHA activation are insufficient to kill CQ resistant parasites.

Relative to PPQ and MFQ, CQ appears to have the greatest ability to render heme inert. We were surprised to observe that PPQ inhibited H-FluNox fluorescence only half as well as CQ in solution. This suggesting that there may be important differences in how these 4-aminoquinolines bind heme. In Dd2 parasites, CQ also inhibited Ac-H-FluNox fluorescence to a greater extent than PPQ at 10 μ M concentrations. Given that the ability of CQ to “defuse” heme in parasites is dependent on PfCRT genotype, it is possible that these results may differ in a PPQ-resistant parasite. In solution, equimolar concentrations of CQ and MFQ inhibited H-FluNox fluorescence to a similar extent, while MFQ had a minimal effect on Ac-H-FluNox fluorescence at even super-pharmacological concentrations. This was somewhat surprising given that MFQ is believed to act in the parasite cytoplasm^{97,98}. It possible that differences in how MFQ binds to heme⁷⁰, affinity of MFQ for other biomolecules⁹⁹, and/or modification of MFQ within the parasite could contribute to this discrepancy.

In the early 2000's, the rise of CQ resistance prompted a search for new combination therapies¹⁰⁰. Several clinical trials were conducted in West Africa to compare efficacy of CQ versus CQ-AS. On day 28, two studies reported similar failure rates for CQ and CQ-AS^{101,102}. In in a third study, cure rates were significantly higher for CQ-AS compared to CQ alone¹⁰³. However, all three studies recommended against use of CQ-AS, due to poor treatment efficacy. Notably, AS-SP and AS-LM were reported to show > 97% efficacy on West African populations during the same time period^{104,105}. For all currently used ACTs, treatment failure appears to be dependent on parasites acquiring resistance to both artemisinins and partner drugs. It is difficult to speculate if

drug-drug interactions could have played a role in resistance evolution. Nevertheless, it remains questionable whether use of DHA-PPQ or AS-ADQ should be enthusiastically supported, given that PPQ and ADQ appear to directly block artemisinin activation.

With the absence of new antimalarials ready for clinical implementation, novel strategies with existing drugs have been proposed to overcome current problems with ACT treatment failures. One such strategy, TACTs, has gained the most traction. For traditional ACTs, the short one hour half-life of artemisinins means that long-lived partner drugs are essentially acting as monotherapies for most of the initial 3-day treatment regimen and a substantial period thereafter. In the case of artemisinin-resistant parasites, partner drugs must clear a significantly higher parasite biomass, which increases the likelihood for resistance to emerge. In theory, TACTs are designed to overcome this flaw, whereby two partner drugs with similar half-lives are used, ensuring that parasites are always exposed to at least two antimalarials ^{106,107}. In clinical trials, DHA-PPQ-MFQ showed improved efficacy over DHA-PPQ in regions of high PPQ-resistance ⁴⁷. However, some have speculated that this was solely due to application of MFQ to MFQ-sensitive parasites and that the TACT would be no more effective than DHA-MFQ ¹⁰⁸. PPQ and MFQ were initially promoted as ideal partner drugs for TACTs as it was thought that either due to collateral sensitivity or fitness costs, dual PPQ and MFQ resistance was unlikely to occur ^{109,110}. Use of edited isogenic lines suggests that PPQ resistance conferring mutations in PfCRT have no effect on MFQ sensitivity, while increased MDR1 copy number, which underlies MFQ-resistance, has no effect on PPQ sensitivity ⁷⁸. Further, more recent genetic surveillance studies of Cambodian parasites cast doubt on the hypothesis that a DHA-PPQ-MFQ resistant parasite would be unfit to spread clinically ¹¹¹. Importantly, we observed superantagonism between PPQ and MFQ in trophozoite-stage isobolograms. This is consistent with previous reports indicating that PPQ and MFQ are antagonistic as assessed with standard 48 or 72 h assays ^{52,55}. The double antagonism between PPQ-DHA and MFQ-PPQ likely explains why we observed no improvement in efficacy between DHA-MFQ and DHA-MFQ-PPQ treatments (**Figure 4a and b**). With the interplay of existing PPQ-resistance, PPQ-MFQ superantagonism, and any potential parasites fitness cost, it is difficult to estimate how quickly DHA-PPQ-MFQ resistant parasites would emerge and spread. Nevertheless, these data collectively indicate that PPQ and MFQ should be scrutinized as partners for TACTs.

The synthetic ozonide, OZ439, was previously poised to replace artemisinins in combination therapies. While OZ439 is believed to have a mechanism of action similar to that of artemisinins, it has a notably improved 46-62 h half-life ⁸⁶ and was shown to be effective against artemisinin-resistant parasites ¹¹²⁻¹¹⁴. While OZ439 showed good efficacy alone ⁸⁶, high treatment failure as part of single-dose combination therapies ultimately resulted in loss of interest of this antimalarial ⁸⁷⁻⁸⁹. However, our data suggest that unsuitable choices in partner drugs may be partly responsible for the poor therapeutic efficacy of these combinations. Both PPQ and FQ rendered heme inert, directly blocking OZ439 activation. Biochemical studies have predicted that due to its less basic behavior, FQ might have a lower propensity to accumulate in the parasite digestive vacuole compared to CQ ⁹⁰. Greater accumulation of FQ in the parasite cytoplasm compared to other quinolines might explain the ability of FQ to greatly diminish Ac-H-FluNox signal even at nanomolar concentrations. Clinically, peak levels of FQ have been reported to reach approximately 0.5 μ M ¹¹⁵, which could significantly impact OZ439 activation. However, unlike CQ, mid-nanomolar FQ concentrations are lethal for parasites which may explain why we did not observe a super antagonistic phenotype for FQ-OZ439.

The antimalarials that we have are a precious resource, and it is only a matter of time before they fail. Our work clearly demonstrates the importance of detailed understanding of drug-drug interactions at the parasite level. While we believe the pulsing assay used in this study better mimics artemisinin pharmacokinetics compared to standard 72 h assays, it is not suitable for high throughput format. Recently, a high throughput assay was developed to evaluate drug-drug and drug-drug-drug interactions in *P. falciparum* ⁵⁵. Using this assay, the authors were able to

evaluate interactions of 16 potential TACT combinations against 13 clinical isolates ⁵⁵[55](#). Similar to our findings, the authors reported that interactions can be highly dependent on parasite genetics. This high throughput assay will undoubtedly be an important first step in evaluating interactions of new combinations. Nevertheless, since this assay relies on a 72 h format, additive interactions should be interpreted with caution, as this format can mask drug-drug interactions ⁵⁸[58](#). Accordingly, pulsing assays like the one used in this study will be an important step for investigating drug-drug interactions, prior to in vivo studies ¹¹⁶[116](#).

Materials and Methods

Reagents

All reagents were purchased from Sigma-Aldrich unless otherwise indicated.

Parasite Culture

Dd2 PfCRT^{Dd2}, Dd2PfCRT^{3D7}, Dd2 PfCRT^{M343L}, and Dd2 K13^{R539T} were kindly provided by Dr. David Fidock (Columbia University) ³⁵[35](#), ⁶¹[61](#). MRA-1284 was obtained through BEI Resources, NIAID, NIH: *Plasmodium falciparum*, Strain IPC_6261, MRA-1284, contributed by Didier Ménard. Parasites were propagated in human red blood cells obtained from St. Louis Children's Hospital blood bank. Cultures were grown at 5% hematocrit in media containing RPMI1640 (Gibco) supplemented with 0.25% (wt/vol) Albumax (Gibco), 15 mg/L hypoxanthine, 110 mg/L sodium pyruvate, 1.19 g/L HEPES, 2.52 g/L sodium bicarbonate, 2 g/L glucose, and 10 mg/L gentamicin and were maintained in air-tight chambers filled with 5% O₂/5%CO₂/90%N₂ at 37°C.

Trophozoite-Stage Dose-Response Assays and Isobolograms

Trophozoite-stage assays were performed as previously described with minor modifications ¹¹⁷[117](#). Briefly, parasites were synchronized to the trophozoites stage using two consecutive treatments with 5% sorbitol 10 h apart, followed by culturing for an additional 14 h. Trophozoite stages were seeded at 0.2% parasitemia and 1% hematocrit in round-bottom 96-well plates. Following 4 h incubation with antimalarials, infected red blood cells were washed a minimum of 5 times for plates containing OZ439, and 4 times for all other antimalarials. Plates were centrifuged, 190 µL of media was carefully removed, and 190 µL of fresh media was added. Following the last wash, cultures were transferred to a new flat-bottom 96-well plate. After 72 hr, parasitemia was assessed by flow cytometry. Cultures were resuspended and 10 µL was transferred to a new 96-well plate containing 50 µL per well of 1 x PBS with 1 x SYBR Green I and 100 nM MitoTracker Deep Red. Following incubation at 37°C for 1 h, plates were read on a BD FACSCanto with a BD HTS automated plate reader and analyzed with BD FACSDiva Software version 8. Approximately 10,000 events were analyzed per well. GraphPad version 10 was used to calculate IC₅₀ values by nonlinear regression.

For dose response assays with fixed concentrations of partner drugs, a 96-well plate was prepared containing 2-fold serial dilutions of DHA media at 100 µL per well. A bulk intermediate culture of parasites was prepared at 2% hematocrit. Partner drugs were added to this intermediate culture at 2x concentrations. After mixing, 100 µL of the intermediate culture was added to the 96-well plate containing DHA serial dilutions, for a final hematocrit of 1% and a 1 x fixed partner drug concentration.

For isobolograms, the following fixed ratios were prepared as starting concentrations: 1:0, 4:1, 2:1, 1:1, 1:2, 1:4, 0:1. The starting concentration of each drug alone (1:0 and 0:1 ratios) represents an approximate 16x IC₅₀ concentration. Trophozoite stages were exposed to 2-fold dilutions of these ratios. Fractional IC₅₀ (FIC₅₀) values were calculated as follows:

$$\text{FIC}_{50} = \frac{\text{IC}_{50} \text{ of drugs in combination}}{\text{IC}_{50} \text{ of individual drug}}$$

FIC₅₀ values were plotted on an XY graph. Mean ΣFIC₅₀ values were calculated by averaging the sums of FIC₅₀ values from 4:1, 2:1, 1:1, 1:2, and 1:4 ratios. Trophozoite-stage IC₅₀ values for Figure 1-figure supplement 1 and Figure 4-figure supplement 2 were obtained from 1:0 or 0:1 ratios. A one-way ANOVA with a Turkey's or Dunnett's multiple comparison test were used to assess statistical significance for IC₅₀ values.

RSAs and Early Ring Stage Dose Response Assays

0-3 hours post invasion (hpi) rings were obtained as previously described¹¹⁸. Briefly, schizonts were enriched using MACS LD magnet columns (Miltenyi Biotec). Schizont enriched cultures were diluted to 2% hematocrit in 3-4 mL of media and incubated for 3 h with shaking. A 5% sorbitol treatment was then used to remove remaining mature parasites. Early ring stages were seeded at 1% parasitemia and 1% hematocrit in round-bottom 96-well plates. 2-fold serial dilutions were prepared starting at 1400 nM DHA, so that both IC₅₀ values and RSA values could be obtained. After incubation for 3 h, antimalarials were washed out and parasitemia was assessed 72 hr later by flow cytometry as described in the previous section. GraphPad Prism version 10 was used to calculate IC₅₀ values by nonlinear regression. RSA survival was obtained by dividing the parasitemia of cultures treated with 700 nM DHA by the untreated control. A one-way ANOVA with a Dunnett's multiple comparison test was used to determine statistical significance for IC₅₀ values and RSA survival values.

Detection of Free Heme In Vitro

H-FluNox was kindly provided by Dr. Tasuku Hirayama (Gifu Pharmaceutical University, Lab of Pharmaceutical & Medicinal Chemistry). Reactions were carried out in a buffer containing 50 mM HEPES (pH 7.4) and 100 μM reduced glutathione. Glutathione was used to convert hemin into heme. Hemin was made up in reaction buffer and 2-fold serial dilutions were performed to obtain a range of heme concentrations starting at 500 nM. H-FluNox was used at a final concentration of 500 nM. Reactions were performed in black-walled clear-bottom 96-well plates. Following incubation with H-FluNox and indicated antimalarials at room temperature for 30 min, fluorescence was read at 485/535 nm ex/em using an EnVision Multimode Plate Reader.

Detection of Free Heme In Live Parasites

Ac-H-FluNox was kindly provided by Dr. Tasuku Hirayama (Gifu Pharmaceutical University, Lab of Pharmaceutical & Medicinal Chemistry). Parasites were synchronized to the trophozoite stage as described above. Cultures were then treated with the indicated compound for the indicated time. Media was removed and parasite cultures were resuspended and incubated in 1 x PBS containing 10 μM Ac-H-FluNox for 30 min at 37°C. Ac-H-FluNox is the cell permeable analog of H-FluNox and is converted to H-FluNox in cells⁷⁴. Following incubation, cultures were washed with 1x PBS and then imaged using a 63x objective on a Zeiss Imager M2 Plus Widefield fluorescence microscope with AxioVision 4.8 software. Except for Figure 3-figure supplement 1b, a fixed exposure of 350 ms was used to capture H-FluNox fluorescence at 470/509 nm ex/em. Exposure for Figure 3-figure supplement 1b was adjusted for each condition so that DV staining could be optimally visualized. For quantification in **Figure 3e, 3f, 4i**, and Figure 3-figure supplement 2, at least ten fields were captured per treatment condition and then mean fluorescence intensity of parasite minus background was quantified with ImageJ. Statistical significance of ≥ 70 parasites from at least three independent drug treatments was assessed using a one-way ANOVA.

Western Blot

Parasite drug treatment, protein harvest, sample preparation, SDS-PAGE, and wet transfer were performed as previously described⁸². Blots were blocked with 3% BSA in TBS-T and incubated overnight with primary antibodies used at 1:1000 in 1x TBS-T. Anti K48-linked ubiquitin was obtained from Cell Signaling Technologies (catalog number: D9D5, rabbit). Anti-PMV was obtained from¹¹⁹. Blots were incubated for 1 h at room temperature with IRDye conjugated goat secondary antibodies (LICOR) at 1:10,000 dilutions. Blots were imaged using a Bio Rad imager with Image Lab Touch Software and then quantified with ImageJ. Significant differences between control and antimalarial treated parasites were determined using a one-way ANOVA with a Dunnett's multiple comparison test.

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Additional information

Contributions

This project was conceptualized by M.R.R. and D.E.G. Experimental methodology, investigation, validation, formal analysis, visualization, and writing of the original draft was done by M.R.R. Manuscript review and editing was performed by M.R.R. and D.E.G. Funding was acquired by D.E.G. and M.R.R. Supervision was provided by D.E.G.

Additional files

Supplemental Material

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Reviewer #1 (Public review):

Summary:

This study set out to investigate potential pharmacological drug-drug interactions between the two most common antimalarial classes, the artemisinins and quinolines. There is a strong rationale for this aim, because drugs from these classes are already widely used in Artemisinin Combination Therapies (ACTs) in the clinic, and drug combinations are an important consideration in the development of new medicines. Furthermore, whilst there is ample literature proposing many diverse mechanisms of action and resistance for the artemisinins and quinolines, it is generally accepted that the mechanisms for both classes involve heme metabolism in the parasite, and that artemisinin activity is dependent on activation by reduced heme. The study was designed to measure drug-drug interactions associated with a short pulse exposure (4 h) that is reminiscent of the short duration of artemisinin exposure obtained after *in vivo* dosing. Clear antagonism was observed between dihydroartemisinin (DHA) and chloroquine, which became even more extensive in chloroquine-resistant parasites. Antagonism was also observed in this assay for the more clinically-relevant ACT partner drugs piperazine and amodiaquine, but not for other ACT partners mefloquine and lumefantrine, which don't share the 4-aminoquinoline structure or mode of action. Interestingly, chloroquine induced an artemisinin resistance phenotype in the standard *in vitro* Ring-stage Survival Assay, whereas this effect was not apparent for piperazine.

The authors also utilised a heme-reactive probe to demonstrate that the 4-aminoquinolines can inhibit heme-mediated activation of the probe within parasites, which suggests that the mechanism of antagonism involves the inactivation of heme, rendering it unable to activate the artemisinins. Measurement of protein ubiquitination showed reduced DHA-induced protein damage in the presence of chloroquine, which is also consistent with decreased heme-mediated activation, and/or with decreased DHA activity more generally.

Overall, the study clearly demonstrates a mechanistic antagonism between DHA and 4-aminoquinoline antimalarials *in vitro*. It is interesting that this combination is successfully used to treat millions of malaria cases every year, which may raise questions about the clinical relevance of this finding. However, the conclusions in this paper are supported by multiple lines of evidence, and the data are clearly and transparently presented, leaving no doubt that DHA activity is compromised by the presence of chloroquine *in vitro*. It is perhaps fortunate that the clinical dosing regimens of 4-aminoquinoline-based ACTs have been sufficient to maintain clinical efficacy despite the non-optimal combination. Nevertheless, optimisation of antimalarial combinations and dosing regimens is becoming more important in the current era of increasing resistance to artemisinins and 4-aminoquinolines. Therefore, these findings should be considered when proposing new treatment regimens (including Tripe-ACTs) and the assays described in this study should be performed on new drug combinations that are proposed for new or existing antimalarial medicines.

Strengths:

This manuscript is clearly written, and the data presented are clear and complete. The key conclusions are supported by multiple lines of evidence, and most findings are replicated with multiple drugs within a class, and across multiple parasite strains, thus providing more confidence in the generalisability of these findings across the 4-aminoquinoline and peroxide drug classes.

A key strength of this study was the focus on short pulse exposures to DHA (4 h in trophs and 3 h in rings), which is relevant to the *in vivo* exposure of artemisinins. Artemisinin resistance has had a significant impact on treatment outcomes in South-East Asia, and is now emerging in Africa, but is not detected using a 'standard' 48 or 72 h *in vitro* growth inhibition assay. It is only in the RSA (a short pulse of 3-6 h treatment of early ring stage parasites) that the resistance phenotype can be detected *in vitro*. Therefore, assays based on this short pulse exposure provide the most relevant approach to determine whether drug-drug interactions are likely to have a clinically relevant impact on DHA activity. These assays clearly showed antagonism between DHA and 4-aminoquinolines (chloroquine, piperaquine, amodiaquine, and ferroquine) in trophozoite stages. Interestingly, whilst chloroquine clearly induced an artemisinin-resistant phenotype in the RSA, piperaquine did not appear to impact the early ring stage activity of DHA, which may be fortunate considering that piperaquine is a currently recommended DHA partner drug in ACTs, whereas chloroquine is not!

The evaluation of additional drug combinations at the end of this paper is a valuable addition, which increases the potential impact of this work. The finding of antagonism between piperaquine and OZ439 in trophozoites is consistent with the general interactions observed between peroxides and 4-aminoquinolines, and it would be interesting to see whether piperaquine impacts the ring-stage activity of OZ439.

The evaluation of reactive heme in parasites using a fluorescent sensor, combined with the measurement of K48-linked ubiquitin, further supports the findings of this study, providing independent read-outs for the chloroquine-induced antagonism.

The in-depth discussion of the interpretation and implications of the results is an additional strength of this manuscript. Whilst the discussion section is rather lengthy, there are important caveats to the interpretation of some of these results, and clear relevance to the future management of malaria that require these detailed explanations.

Overall, this is a high-quality manuscript describing an important study that has implications for the selection of antimalarial combinations for new and existing malaria medicines.

Weaknesses:

This study is an *in vitro* study of parasite cultures, and therefore, caution should be taken when applying these findings to decisions about clinical combinations. The drug concentrations and exposure durations in these assays are intended to represent clinically relevant exposures, although it is recognised that the *in vitro* system is somewhat simplified and there may be additional factors that influence *in vivo* activity. I think this is reasonably well acknowledged in the manuscript.

It is also important to recognise that the majority of the key findings regarding antagonism are based on trophozoite-stage parasites, and one must show caution when generalising these findings to other stages or scenarios. For example, piperaquine showed clear antagonism in trophozoite stages, but not in ring stages under these assay conditions.

The key weakness in this manuscript is the over-interpretation of the mechanistic studies that implicate heme-mediated artemisinin activation as the mechanism underpinning antagonism by chloroquine. In particular, the manuscript title focuses on heme-mediated activation of artemisinins, but this study did not directly measure the activation of artemisinins. The data obtained from the activation of the fluorescent probe are generally supportive of chloroquine suppressing the heme-mediated activation of artemisinins, and I think this is the most likely explanation, but there are significant caveats that undermine this conclusion. Primarily, the inconsistency between the fluorescence profile in the chemical reactions and the cell-based assay raises questions about the accuracy of this readout. In the chemical reaction, mefloquine and chloroquine showed identical inhibition of fluorescence, whereas

piperaquine had minimal impact. On the contrary, in the cell, chloroquine and piperaquine had similar impacts on fluorescence, but mefloquine had minimal impact. This inconsistency indicates that the cellular fluorescence based on this sensor does not give a simple direct readout of the reactivity of ferrous heme, and therefore, these results should be interpreted with caution. Indeed, the correlation between fluorescence and antagonism for the tested drugs is a correlation, not causation. There could be several reasons for the disconnect between the chemical and biological results, either via additional mechanisms that quench fluorescence, or the presence of biomolecules that alter the oxidation state or coordination chemistry of heme or other potential catalysts of this sensor. It is possible that another factor that influences the H-FluNox fluorescence in cells also influences the DHA activity in cells, leading to the correlation with activity. It should be noted that H-FluNox is not a chemical analogue of artemisinins. Its activation relies on Fenton-like chemistry, but with an N-O rather than O-O bond, and it possesses very different steric and electronic substituents around the reactive centre, which are known to alter reactivity to different iron sources. Despite these limitations, the authors have provided reasonable justification for the use of this probe to directly visualise heme reactivity in cells, and the results are still informative, but additional caution should be provided in the interpretation, and the results are not conclusive enough to justify the current title of the paper.

Another interesting finding that was not elaborated by the authors is the impact of chloroquine on the DHA dose-response curves from the ring stage assays. Detection of artemisinin resistance in the RSA generally focuses on the % survival at high DHA concentrations (700 nM) as there is minimal shift in the IC₅₀ (see Figure 2), however, chloroquine clearly induces a shift in the IC₅₀ (~5-fold), where the whole curve is shifted to the right, whereas the increase in % survival is relatively small. This different profile suggests that the mechanism of chloroquine-induced antagonism is different from the mechanism of artemisinin resistance. Current evidence regarding the mechanism of artemisinin resistance generally points towards decreased heme-mediated drug activation due to a decrease in hemoglobin uptake, which should be analogous to the decrease in heme-mediated drug activation caused by chloroquine. However, these different dose-response curves suggest different mechanisms are primarily responsible. Additional mechanisms have been proposed for artemisinin resistance, involving redox or heat stress responses, proteostatic responses, mitochondrial function, dormancy, and PI3K signaling, among others. Whilst the H-FluNox probe generally supports the idea that chloroquine suppresses heme-mediated DHA activation, it remains plausible that chloroquine could induce these, or other, cellular responses that suppress DHA activity.

The other potential weakness in the current manuscript is the interpretation of the OZ439 clinical data. Whilst the observed interaction with piperaquine and ferroquine may have been a contributing factor, it should also be recognised that the low pharmacokinetic exposure in these studies was the primary reason for treatment failure (Macintyre 2017).

Impact:

This study has important implications for the selection of drugs to form combinations for the treatment of malaria. The overall findings of antagonism between peroxide antimalarials and 4-aminoquinolines in the trophozoite stage are robust, and this carries across to the ring stage for chloroquine (but not piperaquine).

The manuscript also provides a plausible mechanism to explain the antagonism, although future work will be required to further explore the details of this mechanism and to rule out alternative factors that may contribute.

Overall, this is an important contribution to the field and provides a clear justification for the evaluation of potential drug combinations in relevant in vitro assays before clinical testing.

Reviewer #2 (Public review):**Summary:**

This manuscript by Rosenthal and Goldberg investigates interactions between artemisinins and their quinoline partner drugs currently used for treating uncomplicated *Plasmodium falciparum* malaria. The authors show that chloroquine (CQ), piperaquine, and amodiaquine antagonize dihydroartemisinin (DHA) activity, and in CQ-resistant parasites, the interaction is described as "superantagonism," linked to the *pfcr1* genotype. Mechanistically, application of the heme-reactive probe H-FluNox indicates that quinolines render cytosolic heme chemically inert, thereby reducing peroxide activation. The work is further extended to triple ACTs and ozonide-quinoline combinations, with implications for artemisinin-based combination therapy (ACT) design, including triple ACTs.

Strengths:

The manuscript is clearly written, methodologically careful, and addresses a clinically relevant question. The pulsing assay format more accurately models *in vivo* artemisinin exposure than conventional 72-hour assays, and the use of H-FluNox and Ac-H-FluNox probes provides mechanistic depth by distinguishing chemically active versus inert heme. These elements represent important refinements beyond prior studies, adding nuance to our understanding of artemisinin-quinoline interactions.

Weaknesses:

Several points warrant consideration. The novelty of the work is somewhat incremental, as antagonism between artemisinins and quinolines is well established. Multiple prior studies using standard fixed-ratio isobologram assays have shown that DHA exhibits indifferent or antagonistic interactions with chloroquine, piperaquine, and amodiaquine (e.g., Davis et al., 2006; Fivelman et al., 2007; Muangnoicharoen et al., 2009), with recent work highlighting the role of parasite genetic background, including *pfcr1* and *pfmdr1*, in modulating these interactions (Eastman et al., 2016). High-throughput drug screens likewise identify quinoline-artemisinin combinations as mostly antagonistic. The present manuscript adds refinement by applying pulsed-exposure assays and heme probes rather than establishing antagonism *de novo*.

The dataset focuses on several parasite lines assayed *in vitro*, so claims about broad clinical implications should be tempered, and the discussion could more clearly address how *in vitro* antagonism may or may not translate to clinical outcomes. The conclusion that artemisinins are predominantly activated in the cytoplasm is intriguing but relies heavily on Ac-H-FluNox data, which may have limitations in accessing the digestive vacuole and should be acknowledged explicitly. The term "superantagonism" is striking but may appear rhetorical; clarifying its reproducibility across replicates and providing a mechanistic definition would strengthen the framing. Finally, some discussion points, such as questioning the clinical utility of DHA-PPQ, should be moderated to better align conclusions with the presented data while acknowledging the complexity of *in vivo* pharmacology and clinical outcomes.

Despite these mild reservations, the data are interesting and of high quality and provide important new information for the field.

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Reviewer #3 (Public review):**Summary:**

The authors present an in vitro evaluation of drug-drug interactions between artemisinins and quinoline antimalarials, as an important aspect for screening the current artemisinin-based combination therapies for *Plasmodium falciparum*. Using a revised pulsing assay, they report antagonism between dihydroartemisinin (DHA) and several quinolines, including chloroquine, piperazine (PPQ), and amodiaquine. This antagonism is increased in CQ-resistant strains in isobologram analyses. Moreover, CQ co-treatment was found to induce artemisinin resistance even in parasites lacking K13 mutations during the ring-stage survival assay. This implies that drug-drug interactions, not just genetic mutations, can influence resistance phenotypes. By using a chemical probe for reactive heme, the authors demonstrate that quinolines inhibit artemisinin activation by rendering cytosolic heme chemically inert, thereby impairing the cytotoxic effects of DHA. The study also observed negative interactions in triple-drug regimens (e.g., DHA-PPQ-Mefloquine) and in combinations involving OZ439, a next-generation peroxide antimalarial. Taken together, these findings raise significant concerns regarding the compatibility of artemisinin and quinoline combinations, which may promote resistance or reduce efficacy.

Throughout the manuscript, no combinations were synergistic, which necessitates comparing the claims to a synergistic combination as a control. The lack of this positive control makes it difficult to contextualize the observed antagonism. Including a known synergistic pair (e.g., artemisinin + lumefantrine) throughout the study would have provided a useful benchmark to assess the relative impact of the drug interactions described.

Strengths:

This study demonstrates the following strengths:

- (1) The use of a pulsed in vitro assay that is more physiologically relevant than the traditional 48h or 72h assays.
- (2) Small molecule probes, H-FluNox, and Ac-H-FluNox to detect reactive cytosolic heme, demonstrating that quinolines render heme inert and thereby block DHA activation.
- (3) Evaluates not only traditional combinations but also triple-drug combinations and next-generation artemisinins like OZ439. This broad scope increases the study's relevance to current treatment strategies and future drug development.
- (4) By using the K13 wild-type parasites, the study suggests that resistance phenotypes can emerge from drug-drug interactions alone, without requiring genetic resistance markers.

Weaknesses:

- (1) No combinations are shown as synergistic: it could be valuable to have a combination that shows synergy as a positive control (e.g., artemisinin + lumefantrine) throughout the manuscript. The absence of a synergistic control combination in the experimental design makes it more challenging to evaluate the relative impact of the described drug interactions.
- (2) Evaluation of the choice of drug-drug interactions: How generalizable are the findings across a broad range of combinations, especially those with varied modes of action?
- (3) The study would also benefit from a characterization of the molecular basis for the observed heme inactivation by quinolines to support this hypothesis - while the probe experiments are valuable, they do not fully elucidate how quinolines specifically alter heme chemistry at the molecular level.

(4) Suggestion of alternative combinations that show synergy could have improved the significance of the work.

(5) All data are derived from in vitro experiments, without accompanying an in vivo validation. While the pulsing assay improves physiological relevance, it still cannot fully capture the complexity of drug pharmacokinetics, host-parasite interactions, or immune responses present in living organisms.

(6) The absence of pharmacokinetic/pharmacodynamic modeling leaves questions about how the observed antagonism would manifest under real-world dosing conditions.

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Author response:

Reviewer #1:

We thank the reviewer for their thoughtful summary of this manuscript. It is important to note that DHA-PPQ did show antagonism in RSAs. In this modified RSA, 200 nM PPQ alone inhibited growth of PPQ-sensitive parasites approximately 20%. If DHA and PPQ were additive, then we would expect that addition of 200 nM PPQ would shift the DHA dose response curve to the left and result in a lower DHA IC₅₀. Please refer to Figure 4a and b as examples of additive relationships in dose-response assays. We observed no significant shift in IC₅₀ values between DHA alone and DHA + PPQ. This suggests antagonism, albeit not to the extent seen with CQ. We will modify the manuscript to emphasize this point. As the reviewer pointed out, it is fortunate that despite being antagonistic, clinically used artemisinin-4-aminoquinoline combinations are effective, provided that parasites are sensitive to the 4-aminoquinoline. It is possible that superantagonism is required to observe a noticeable effect on treatment efficacy (Sutherland et al. 2003 and Kofoed et al. 2003), but that classical antagonism may still have silent consequences. For example, if PPQ blocks some DHA activation, this might result in DHA-PPQ acting more like a pseudo-monotherapy. However, as the reviewer pointed out, while our data suggest that DHA-PPQ and AS-ADQ are “non-optimal” combinations, the clinical consequences of these interactions are unclear. We will modify the manuscript to emphasize the later point.

While the Ac-H-FluNox and ubiquitin data point to a likely mechanism for DHA-quinoline antagonism, we agree that there are other possible mechanisms to explain this interaction. We will temper the title and manuscript to reflect these limitations. Though we tried to measure DHA activation in parasites directly, these attempts were unsuccessful. We acknowledge that the chemistry of DHA and Ac-H-FluNox activation is not identical and that caution should be taken when interpreting these data. Nevertheless, we believe that Ac-H-FluNox is the best currently available tool to measure “active heme” in live parasites and is the best available proxy to assess DHA activation in live parasites. Both in vitro and in parasite studies point to a role for CQ in modulating heme, though an exact mechanism will require further examination. Similar to the reviewer, we were perplexed by the differences observed between in vitro and in parasite assays with PPQ and MFQ. We proposed possible hypotheses to explain these discrepancies in the discussion section. Interestingly, our data correlate well with hemozoin inhibition assays in which all three antimalarials inhibit hemozoin formation in solution, but only CQ and PPQ inhibit hemozoin formation in parasites. In both assays, in-parasite experiments are likely to be more informative for mechanistic assessment.

It remains unclear why K13 genotype influences RSA values, but not early ring DHA IC₅₀ values. In K13^{WT} parasites, both RSA values and DHA IC₅₀ values were increased 3-5 fold

upon addition of CQ. This suggests that CQ-mediated resistance is more robust than that conferred by K13 genotype. However, this does not necessarily suggest a different resistance mechanism. We acknowledge that in addition to modulating heme, it is possible that CQ may enhance DHA survival by promoting parasite stress responses. Future studies will be needed to test this alternative hypothesis. This limitation will be acknowledged in the manuscript. We will also address the reviewer's point that other factors, including poor pharmacokinetic exposure, contributed to OZ439-PPQ treatment failure.

Reviewer #2:

We appreciate the positive feedback. We agree that there have been previous studies, many of which we cited, assessing interactions of these antimalarials. We also acknowledge that previous work, including our own, has shown that parasite genetics can alter drug-drug interactions. We will include the author's recommended citations to the list of references that we cited. Importantly, our work was unique not only for utilizing a pulsing format, but also for revealing a superantagonistic phenotype, assessing interactions in an RSA format, and investigating a mechanism to explain these interactions. We agree with the reviewer that implications from this in vitro work should be cautious, but hope that this work contributes another dimension to critical thinking about drug-drug interactions for future combination therapies. We will modify the manuscript to temper any unintended recommendations or implications.

The reviewer notes that we conclude "artemisinins are predominantly activated in the cytoplasm". We recognize that the site of artemisinin activation is contentious. We were very clear to state that our data combined with others suggest that artemisinins can be activated in the parasite cytoplasm. We did not state that this is the primary site of activation. We were clear to point out that technical limitations may prevent Ac-H-FluNox signal in the digestive vacuole, but determined that low pH alone could not explain the absence of a digestive vacuole signal.

With regard to the "reproducibility" and "mechanistic definition" of superantagonism, we observed what we defined as a one-sided superantagonistic relationship for three different parasites (Dd2, Dd2 PfCRT^{Dd2}, and Dd2 K13^{R539T}) for a total of nine independent replicates. In the text, we define that these isoboles are unique in that they had mean Σ FIC50 values > 2.4 and peak Σ FIC50 values > 4 with points extending upward instead of curving back to the axis. As further evidence of the reproducibility of this relationship, we show that CQ has a significant rescuing effect on parasite survival to DHA as assessed by RSAs and IC50 values in early rings.

Reviewer #3:

We thank the reviewer for their positive feedback. We acknowledge that no combinations tested in this manuscript were synergistic. However, two combinations, DHA-MFQ and DHA-LM, were additive, which provides context for contextualizing antagonistic relationships. We have previously reported synergistic and additive isobolograms for peroxide-proteasome inhibitor combinations using this same pulsing format (Rosenthal and Ng 2021). These published results will be cited in the manuscript.

We believe that these findings are specific to 4-aminoquinoline-peroxide combinations, and that these findings cannot be generalized to antimalarials with different mechanisms of action. Note that the aryl amino alcohols, MFQ and LM, were additive with DHA. Since the mechanism of action of MFQ and LM are poorly understood, it is difficult to speculate on a mechanism underlying these interactions.

We agree with the reviewer that while the heme probe may provide some mechanistic insight to explain DHA-quinoline interactions, there is much more to learn about CQ-heme chemistry, particularly within parasites.

The focus of this manuscript was to add a new dimension to considerations about pairings for combination therapies. It is outside the scope of this manuscript to suggest alternative combinations. However, we agree that synergistic combinations would likely be more strategic clinically.

An in vitro setup allows us to eliminate many confounding variables in order to directly assess the impact of partner drugs on DHA activity. However, we agree that in vivo conditions are incredibly more complex, and explicitly state this.

We agree that in the future, modeling studies could provide insight into how antagonism may contribute to real-world efficacy. This is outside the scope of our studies.

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